

Paul Jialiang Wu, PhD

Research Engineer — Humanoids · Sim-to-Real · World Models

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SUMMARY

Sim-to-real is a calibration problem, and calibration is a measurement problem: my NeurIPS submission cuts transfer error 40% by calibrating the world model online rather than randomizing the domain.

WHY GOOGLE DEEPMIND — RESEARCH ENGINEER, HUMANOIDS

- **Self-Calibrating World Models (NeurIPS 2026, under review):** Online Bayesian calibration for sim-to-real transfer; 40% transfer-error reduction vs domain randomization. (NeurIPS 2026 under review)
- **Physical AI: The Next Frontier in AI and Robotics:** Preprints.org, Apr 2026. DOI: 10.20944/preprints202604.0549.v1. (DOI 10.20944/preprints202604.0549.v1)
- **The eval framework that decided production model selection:** Primary technical interface on an executive-level enterprise AI deployment (Accenture OpenAI-DSI). Built the multi-model evaluation framework comparing 6+ VLMs (Gemini-class, NVIDIA VSS, Cosmos-Predict2) on accuracy, latency under load, edge-case failure modes and integration cost — the basis for production model selection. Shipped a confidence-calibrated selective-verification pipeline routing only high-uncertainty outputs to human review: ~90% annotation cost reduction. (Accenture, Physical AI Team (2024–present))
- **Audited my own benchmark and it demoted my own published result:** A VLM failure taxonomy as data → gated probe suite. Then a 3-repeat variance run found the grader, not the model, was the least reliable component: three harness bugs were MANUFACTURING failures (an empty API response scored 0.0; a truncated answer graded as complete; the suite's own tests never gated by CI). Fixing them cut instability from 5 unstable probe/model pairs to 1, and forced a correction to a capability claim I had already put in a paper table. Every spec change was pre-registered with falsifiable per-probe predictions BEFORE the run that tested it. (github.com/wjlgatech/FM-os/tree/main/skills/vlm-failure-probe)
- **A training API's primitives, keyless, with a forgetting gate:** Four primitives + LoRA implemented keyless in numpy; a forgetting gate catches +208% catastrophic forgetting and rolls the naive recipe back. Certified 98/100. (github.com/wjlgatech/FM-os/tree/main/skills/tinker-loop)

ALSO SHIPPED

- **FM-os — spec-as-data FM-ops hub (creator)** — github.com/wjlgatech/FM-os
- **Self-improving harness with an independent referee** — github.com/wjlgatech/loop-engineering-anything

EXPERIENCE

Applied AI Scientist / Forward Deployed Engineer — Accenture, Physical AI Team · Jan 2024–Present

- Primary technical interface for an executive-level enterprise AI deployment (OpenAI-DSI engagement); recognized for a "no surprises" standard built on deep model understanding and contextual judgment at decision points.
- Built the multi-model evaluation framework comparing 6+ VLMs (Gemini-class, NVIDIA VSS, Cosmos-Predict2) on accuracy, latency under load, edge-case failure modes and integration cost — the basis for production model selection.

- Shipped a confidence-calibrated selective-verification pipeline routing only high-uncertainty outputs to human review: ~90% annotation cost reduction.
- Technical lead on 4 simultaneous production systems: synthetic data generation, zero-shot safety detection, adaptive evaluation under distribution shift, vision-language reasoning over live sensor streams.

Principal Data Scientist — Genentech · Jun 2021–Dec 2022

- Production ML for biomedical research; 3 open-source PyPI packages for multimodal learning; 85% reduction in unproductive process time.

Principal Data Scientist — Galvanize Inc. · Sep 2019–Jun 2021

- Technical lead across 13+ end-to-end ML systems for Fortune 500 clients; 5-star-rated delivery.

PUBLICATIONS

- **Self-Calibrating World Models (NeurIPS 2026, under review)** — Online Bayesian calibration for sim-to-real transfer; 40% transfer-error reduction vs domain randomization.
- **Physical AI: The Next Frontier in AI and Robotics** — Preprints.org, Apr 2026. DOI: 10.20944/preprints202604.0549.v1.

TECHNICAL SKILLS

Tooluse: MCP servers/benchmarks, agent harness design, evidence-gated trace verification, instruction-following probes, connector generation (OpenAPI→CLI/MCP)

Posttraining: DPO from scratch (torch+peft), LoRA/QLoRA, preference-data pipelines from eval failures, reward-hack detection, win-rate gates, ship/rollback

Evaluation: LLM-as-judge referees (maker≠checker), failure-taxonomy-as-data suites, held-out gated benchmarks, pre-registered criteria, variance reporting, confidence calibration, distribution-shift detection

Engineering: Python, PyTorch, numpy, JAX-adjacent, FastAPI, Docker/K8s, CI/CD, large-codebase debugging (543-test OSS), Go, SQLite/Postgres

EDUCATION

Yale University — National BioMed Fellow, Computational Immunology · **Georgia Tech** — PhD, Bioinformatics · **University of South Carolina** — MS, Mathematics & CS · **Sun Yat-Sen University** — BS, Applied Mathematics

THE EDGE I'D BE LEARNING, NOT TEACHING

My robotics work is world-model and perception-side; I have not shipped whole-body-control or locomotion policies on hardware.